PRODUCT REQUIREMENTS DOCUMENT

Photo Puzzle Game

A multiplayer, turn-based photo guessing game

Version 1.0 · March 2026

Status: Ready for Development

Overview

This PRD defines a browser-based, mobile-first multiplayer photo puzzle game. Players take turns setting photo-based puzzles for each other and guessing each other's puzzles, with one turn per day. The game is designed for small invite-based groups with open-ended play and cumulative scoring.

Core Concept

Each day constitutes one turn. Every player in a group simultaneously acts as a setter (creating one puzzle for the others) and a guesser (attempting to solve each other player's puzzle). For a group of N players, there are N concurrent rounds running at any time.

A setter uploads a personal photograph and provides metadata (What, Who, Where, When). They can optionally add visual and textual clues. Guessers see a limited view of the photo and attempt to identify the metadata. Correct answers lock in points; incorrect answers require the guesser to either reveal a clue (reducing available points) or concede.

Key Properties

- **Invite-based groups:** Private games created and joined by invitation
- **Asynchronous:** Players set and guess independently within a 24-hour window
- **Open-ended:** No fixed end date; cumulative leaderboard
- **Mobile-first:** Designed primarily for mobile browsers with desktop support
- **AI-augmented:** LLM-powered clue generation and answer evaluation

Gameplay Mechanics

Setting a Puzzle

The setter flow is a 6-step wizard:

1. **Upload Subject Photo:** The setter selects a photo from their device. On mobile, this uses the native photo picker. On desktop, a standard file upload dialog. This is the core image of the puzzle.
2. **Define Sub-portions (optional):** The setter can draw up to 3 rectangles on the subject photo. These create cropped views that are revealed progressively to the guesser. The first sub-portion is shown for free as the initial view.
3. **Upload Clue Photos (optional):** Up to 3 additional photographs that serve as visual hints. These are separate images, not crops of the subject photo.
4. **Enter Metadata:** The setter provides answers for four fields. If a field is not applicable (e.g. no people in the photo), the setter marks it as N/A. Fields marked N/A are excluded from the guessing round and scoring.
5. **Add Text Clues (optional):** The setter can request AI-generated clues (the system produces 5, the setter picks 0–3) and/or manually write up to 3 clues. Total across AI-selected and manual cannot exceed 3.
6. **Review & Submit:** The setter sees a summary including the total points available, all reveals, and the answers. They confirm and submit.

Metadata Fields

Field	Description	Matching Rules
What	What is it a photo of? (e.g. "Tower Bridge")	Fuzzy LLM matching. Synonyms and reasonable descriptions accepted.
Who	Who is in the picture? Person or named animal. Mark N/A if nobody.	Name matching including common nicknames (Jess/Jessica, Mike/Michael).
Where	Where was it taken? (e.g. "London")	Geographic containment. "North London" matches "London". LLM evaluates equivalence.
When	When was it taken? Year only.	Exact year match only.

EXIF extraction: If the subject photo contains EXIF metadata, the system should attempt to pre-fill Where (GPS coordinates reverse-geocoded) and When (date taken). The setter can override or accept these values.

AI Clue Generation

When the setter taps the “Generate AI Clues” button, the system sends the subject photo to a vision-capable LLM with a prompt requesting 5 varied, progressively helpful clues. The clues should:

- Avoid directly naming the subject, location, or people
- Range from oblique hints to more direct ones
- Reference visual, historical, cultural, or geographic attributes
- Be phrased as single sentences

The setter selects 0 to 3 of the 5 generated clues. They can also write their own, up to a combined total of 3 text clues.

Guessing a Puzzle

When a guesser opens a puzzle, they see the initial image:

- **If sub-portions exist:** The first sub-portion is displayed, with the rest of the subject photo blurred out. This initial sub-portion is free (does not cost a reveal).
- **If no sub-portions:** The full subject photo is displayed.

The guesser enters answers for all non-N/A fields and submits. Each field is evaluated independently:

- **Correct:** The field locks in immediately with points awarded at the current reveal level. It is shown as solved and cannot be changed.
- **Incorrect:** The field remains unsolved. The guesser must choose to either reveal one additional piece of information (any type) or concede.

One guess per field per reveal level. The guesser cannot re-submit answers for the same field at the same reveal level. After submitting, if any fields are wrong, they must reveal a clue or concede before guessing again.

Reveal choice: When revealing, the guesser chooses the type (next sub-portion, a clue photo, or a text clue). The guesser picks which category but not which specific item; reveals are sequential within each type.

Full photo: The full subject photo is only shown if the guesser concedes or runs out of reveals. It is never shown while the game is active (if sub-portions were provided).

Concession

The guesser can concede at any point. On concession:

- Points for any previously correct fields are kept
- Unsolved fields score 0
- The full subject photo is revealed
- All correct answers are shown

Missed Turns

If a guesser does not play by 06:00 the following day (configurable), they score 0 for that puzzle. The puzzle is then resolved and the setter receives maximum points for that guesser.

Scoring System

Points Pool Calculation

The total points available for a puzzle are:

```
Total Points = 25 + (Number of Reveals × 10)
```

Where “Number of Reveals” is the sum of sub-portions + clue photos + text clues.

Example Calculation

Component	Count	Points Added
Base	—	25
Sub-portions	2	20
Clue photos	1	10
Text clues	2	20
Total	**5 reveals**	**75 points**

Guesser Scoring

Points are divided equally across all non-N/A fields. With the example above and 3 non-N/A fields, each field is worth 75 ÷ 3 = 25 points at the first guess level.

Each reveal reduces the total by 10 points:

- First guess (0 reveals used): 75 points total, 25 per field
- After 1 reveal: 65 points total, ~21.7 per field
- After 2 reveals: 55 points total, ~18.3 per field
- After all 5 reveals: 25 points total, ~8.3 per field

A field’s points are locked in at the level when the correct answer is submitted. Different fields can be scored at different reveal levels.

Setter Scoring

The setter earns the inverse of the guesser’s score:

```
Setter Points = Total Points − Guesser Points
```

This incentivises setters to both create difficult puzzles and provide generous clue material. A setter who provides many reveals (increasing the total pool) but whose puzzle remains unsolved earns maximum points. With multiple guessers, the setter earns separate scores against each guesser.

Scoring Edge Cases

Scenario	Guesser Score	Setter Score
Perfect guess (all fields, first try)	Total Points	0
Concession after some correct fields	Sum of locked-in field points	Total − Guesser
Full concession (no correct fields)	0	Total Points
Missed turn (no play by 06:00)	0	Total Points

Cumulative Leaderboard

All points (from both setting and guessing) accumulate over time. The leaderboard shows total points, with breakdowns by setting points and guessing points. There is no end condition; the game continues indefinitely.

Answer Evaluation

All answer evaluation is performed by an LLM to allow for fuzzy, natural-language matching. The system sends the setter's answer and the guesser's answer as a pair, along with the field type, and asks for a binary match/no-match decision.

Matching Rules by Field

What

The broadest matching. Accept synonyms, reasonable descriptions, and alternate names. "Tower Bridge" should match "the bridge with two towers in London". The LLM should evaluate semantic equivalence, not string similarity.

Who

Name matching with awareness of common nicknames, diminutives, and formal/informal variants. Examples:

- Jess ↔ Jessica
- Mike ↔ Michael
- Dave ↔ David
- Bob ↔ Robert
- Full name match if only first name given or vice versa

Where

Geographic containment and equivalence. A more specific answer should match a more general one:

- "North London" matches "London"
- "Manhattan" matches "New York"
- "Tuscany" matches "Italy"
- But "Paris" does not match "France" (too vague — the LLM should use judgement)

When

Year only. Exact numeric match. The LLM should extract the year from both answers and compare. "2019" matches "2019". "Last summer" does not match (must be a year).

LLM Evaluation Prompt

The evaluation prompt should be structured to return a deterministic result. Suggested approach:

- System prompt establishing the role as a quiz answer evaluator
- Include field type (What/Who/Where/When) and corresponding matching rules

- Provide setter’s answer and guesser’s answer
- Request a JSON response: { "match": true/false, "reasoning": "..." }
- Use low temperature (0.0–0.1) for consistency
- Cache identical evaluation pairs to avoid redundant API calls

Design System

The prototypes establish a dark, photo-centric aesthetic optimised for mobile. The design system should be implemented as reusable design tokens and components.

Colour Palette

Token	Value	Usage
bg-primary	#111827	Page background
bg-card	#1F2937	Card / input backgrounds
bg-border	#374151	Borders, dividers
text-primary	#F5F5F4	Primary text (headings, body)
text-secondary	#9CA3AF	Secondary text, labels, hints
text-muted	#6B7280	Muted text, timestamps
accent-primary	#F59E0B	Primary accent (amber-400)
accent-dark	#D97706	Darker accent for gradients (amber-600)
accent-deep	#92400E	Deep accent for gradient endpoints (amber-800)
success	#10B981	Correct answers, positive states
success-bg	#05966922	Success backgrounds (translucent)
error	#EF4444	Wrong answers, negative states
error-bg	#EF444422	Error backgrounds (translucent)
clue-sub	#3B82F6	Sub-portion indicator (blue)
clue-photo	#8B5CF6	Clue photo indicator (purple)
clue-text	#F59E0B	Text clue indicator (amber)

Typography

Element	Font	Size	Weight
Page title	DM Sans	20px	800
Section label	JetBrains Mono	12px	400
Body text	DM Sans	14–15px	400–500

Score display	JetBrains Mono	22px	800
Points notation	JetBrains Mono	12–13px	700
Button text	DM Sans	15px	700
Small labels	DM Sans	11–12px	600
Hint text	DM Sans	12–13px	400

Font loading: Import DM Sans (400, 500, 600, 700, 800) and JetBrains Mono (400, 700) from Google Fonts.

Spacing & Layout

- **Max width:** 480px, centred. All content constrained to mobile viewport.
- **Page padding:** 20px horizontal. 16px vertical for header.
- **Card padding:** 14–16px
- **Component gaps:** 8–12px between items, 16–20px between sections
- **Border radius:** 8px for small elements, 10–12px for cards and images

Component Patterns

Buttons

- **Primary:** Linear gradient 135deg from #D97706 to #F59E0B. Black text, 700 weight. Border-radius 12px, padding 14px.
- **Secondary:** Background #1F2937, border 1px solid #374151. Light text #E5E7EB, 600 weight.
- **Disabled:** Background #374151, text #6B7280, no pointer cursor.
- **Destructive:** Background #EF4444, white text. Used for remove/delete actions.

Inputs

- Background #1F2937, border 1px solid #374151
- Border radius 10px, padding 12px 14px
- Text colour #F5F5F4, font size 15px
- Focus state: border-color changes to #F59E0B
- Placeholder colour: inherits from text-secondary

Status Cards

- **Correct field:** Background #05966922, border 1px solid #05966944. Green text for label, lighter green for value, amber for points.
- **Wrong field:** Background #EF444422, border 1px solid #EF444444. Red text.

- **Revealed clue:** Background #F59E0B11, border 1px solid #F59E0B33. Warm yellow text #FDE68A.

Fixed Bottom Navigation

Sticky bottom bar with gradient fade from transparent to bg-primary. Max-width 480px, centred. Contains primary action button (right, flex: 2) and secondary action (left, flex: 1). Padding 16px 20px.

Fullscreen Image Viewer

Overlay triggered by tapping any image. Features:

- Fixed position, full viewport, background #000000EE, z-index 9999
- Pinch-to-zoom on mobile, scroll-to-zoom on desktop
- Drag to pan when zoomed (clamped to image bounds)
- Double-tap to toggle between 1× and 2.5×
- Close button: 40px circle, top-right, background #000000AA
- Zoom indicator at bottom when scale > 1× (JetBrains Mono, #000000AA pill)

Step Indicator (Setter)

Horizontal step bar with numbered circles (28px, JetBrains Mono). Completed steps show tick and amber fill. Current step highlighted. Connected by 12px lines. Horizontally scrollable on small screens.

Data Model

The following entities are required. Field names are illustrative; use the conventions of your chosen ORM/database.

Group

Field	Type	Notes
id	UUID	Primary key
name	String	Display name for the group
created_at	Timestamp	
invite_code	String (unique)	Used to join the group
turn_deadline_hour	Integer	Hour (24h) when turn expires. Default 6.

User

Field	Type	Notes
id	UUID	Primary key
display_name	String	Shown to other players
email	String	For auth
avatar_url	String (nullable)	Profile image

GroupMembership

Field	Type	Notes
group_id	FK → Group	
user_id	FK → User	
joined_at	Timestamp	
total_score	Integer	Cached cumulative score

Puzzle

Field	Type	Notes
id	UUID	Primary key
group_id	FK → Group	
setter_id	FK → User	
turn_date	Date	Which day this puzzle belongs to
subject_photo_url	String	Stored image URL
sub_portions	JSONB	Array of {x, y, w, h} as percentages
clue_photo_urls	JSONB	Array of stored image URLs
answer_what	String (nullable)	Null if N/A
answer_who	String (nullable)	Null if N/A
answer_where	String (nullable)	Null if N/A
answer_when	String (nullable)	Null if N/A (year as string)
text_clues	JSONB	Array of clue strings
total_points	Integer	Calculated: 25 + reveals × 10
created_at	Timestamp	

GuessSession

Field	Type	Notes
id	UUID	Primary key
puzzle_id	FK → Puzzle	
guesser_id	FK → User	
status	Enum	active | completed | conceded | expired
reveals_used	Integer	Count of reveals consumed
reveal_history	JSONB	Ordered array of reveal types chosen
score	Integer	Guesser’s final score
setter_score	Integer	Total − guesser score
completed_at	Timestamp (nullable)	

Guess

Field	Type	Notes
id	UUID	Primary key
session_id	FK → GuessSession	
field	Enum	what | who | where | when
value	String	The guesser’s answer
reveal_level	Integer	How many reveals had been used
is_correct	Boolean	Result of LLM evaluation
points_awarded	Integer	0 if incorrect
created_at	Timestamp	

Technical Architecture

Recommended Stack

Layer	Technology	Rationale
Frontend	Next.js (App Router) + React	SSR for initial load, client-side interactivity for game UI
Styling	Tailwind CSS	Utility-first matches the token-based design system
Database	Supabase (PostgreSQL)	Auth, storage, real-time subscriptions, RLS
Image Storage	Supabase Storage	Signed URLs, image transformations
Auth	Supabase Auth	Magic link or social login
LLM (evaluation)	Anthropic Claude API	Answer matching, temperature 0.0–0.1
LLM (clue gen)	Anthropic Claude API (vision)	Clue generation from subject photo
Background Jobs	Supabase Edge Functions or pg_cron	Turn expiry, score rollup
Hosting	Vercel	Pairs naturally with Next.js

API Endpoints

Key endpoints (RESTful or tRPC):

Method	Path	Description
POST	/api/groups	Create a new group
POST	/api/groups/:id/join	Join via invite code
POST	/api/puzzles	Submit a puzzle (multipart: images + metadata)
GET	/api/puzzles/:id/play	Get puzzle for guessing (respects reveal state)
POST	/api/puzzles/:id/guess	Submit guesses for current reveal level

POST	/api/puzzles/:id/reveal	Consume a reveal (specify type)
POST	/api/puzzles/:id/concede	Concede the puzzle
POST	/api/puzzles/:id/generate-clues	Generate AI clues from subject photo
GET	/api/groups/:id/leaderboard	Cumulative scores
GET	/api/groups/:id/activity	Setter view of guesser activity

Image Handling

- **Upload:** Subject photos and clue photos uploaded to Supabase Storage with signed URLs. Strip EXIF on upload (except GPS and date for pre-fill) for privacy.
- **Sub-portions:** Stored as percentage-based coordinates. The frontend renders the crop at display time. No server-side image processing required.
- **Transformation:** Use Supabase image transformations for thumbnails and optimised delivery. Generate WebP variants.
- **Size limits:** Max 10MB per image. Resize on upload to max 2000px on longest edge.

Real-time Updates

Supabase Realtime for:

- Notifying setter when a guesser submits/reveals/concedes
- Updating leaderboard in real-time
- Alerting users when a new puzzle is available

Background Jobs

- **Turn expiry:** Run daily at the configured deadline hour. Mark all active GuessSessions as “expired” with score 0. Award full points to setters.
- **Score rollup:** After each puzzle completion, update GroupMembership.total_score.

Non-Functional Requirements

Performance

Metric	Target
Initial page load (LCP)	< 2.5 seconds on 4G mobile
Image display (after tap)	< 500ms with lazy loading
Guess submission round-trip	< 2 seconds (including LLM evaluation)
AI clue generation	< 10 seconds (with loading indicator)
Fullscreen image open	< 100ms (no network, pure UI)

Scalability

- Initial target: 100 groups, 2–6 players per group
- LLM API calls should be queued if rate-limited, not dropped
- Image storage scales independently via Supabase Storage
- Database should handle 10,000 puzzles without performance degradation

Security

- **Authentication:** All API endpoints require authenticated user
- **Authorisation:** Row-Level Security in Supabase. Users can only access puzzles in their groups. Guessers cannot see answers or other guessers' activity. Setters can see all guesser activity for their puzzles.
- **Image privacy:** Signed URLs with expiry. EXIF GPS data stripped after extraction. No public image URLs.
- **Answer security:** Answers are never sent to the guesser's client. All evaluation happens server-side via LLM.
- **Input validation:** All user inputs sanitised. Image type and size validated server-side.

Accessibility

- All interactive elements keyboard-navigable
- ARIA labels on image viewers, buttons, and form fields
- Minimum contrast ratio 4.5:1 for body text (the dark theme with amber accents meets this)
- Touch targets minimum 44px × 44px on mobile
- Screen reader announcements for score changes and answer results

Browser Support

- Safari iOS 15+ (primary target)
- Chrome Android 90+
- Chrome/Edge/Firefox desktop (latest 2 versions)
- Progressive enhancement: core functionality without JavaScript for initial render

Offline / Poor Connectivity

- Images lazy-loaded with blur-up placeholders
- Optimistic UI for guess submission (show pending state, reconcile on response)
- Graceful error handling if LLM evaluation fails (retry with backoff, queue for later)
- Service worker for caching static assets (future enhancement)

Observability

- Structured logging for all API endpoints
- LLM evaluation results logged (for tuning matching quality)
- Error tracking (Sentry or equivalent)
- Basic analytics: puzzles set/day, guesses/day, average score, completion rate

Data Retention

- Images retained as long as the group is active
- Puzzle and guess data retained indefinitely for leaderboard integrity
- Users can delete their account; their puzzles become anonymous but remain in the game

UX Flows

First-Time User

1. Land on home screen (not logged in)
2. Sign up / log in (magic link or social)
3. See empty dashboard with “Create Group” or “Join Group”
4. Create group → get invite link to share
5. Or: tap invite link → join group → see dashboard

Daily Loop (Setter)

1. Open app → dashboard shows “Set today’s puzzle” prompt
2. Tap → enter 6-step setter wizard
3. Upload photo → optionally add sub-portions → optionally add clue photos
4. Enter What/Who/Where/When (or N/A)
5. Optionally generate/write clues
6. Review and submit
7. Return to dashboard → see pending guessers

Daily Loop (Guesser)

1. Open app → dashboard shows puzzles to solve
2. Tap a puzzle → see initial image
3. Enter guesses for all fields → submit
4. See results: correct fields lock in (green), wrong fields shown (red)
5. Choose: reveal a clue (pick type) or concede
6. If reveal: see new info, guess again
7. Repeat until all correct, conceded, or out of clues
8. See final score and setter’s score

Setter Activity View

After submitting a puzzle, the setter can monitor guesser activity:

- Which guessers have started, are in progress, or have finished
- For each guesser: their guess history per field, which reveals they chose, their final score
- The setter’s points earned from each guesser
- This view updates in real-time via Supabase Realtime

Future Considerations

The following are explicitly out of scope for V1 but should be considered in the architecture:

- **Push notifications:** Remind players to set/guess before deadline. Architecture should support adding web push or native notifications later.
- **Streaks:** Track consecutive days of play. Display on profile/leaderboard.
- **Photo moderation:** Automated NSFW detection on uploaded images. Flag for review.
- **Puzzle archive:** Browse past puzzles and their solutions. “Replay” mode for new group members.
- **Group settings:** Configurable deadline hour, timezone, weekend skips.
- **Multi-language:** LLM evaluation already handles this implicitly; UI localisation would need separate work.
- **Native apps:** If the PWA approach has limitations (camera access, push), consider React Native or Capacitor wrappers.

Appendix: Reveal Ordering Logic

When the guesser chooses to reveal information, they select a category. Reveals within each category are sequential:

Category	Reveal Order	Visual Treatment
Sub-portions	In order drawn by setter (1, 2, 3)	Blurred photo with sharp cropped area. Border colour #3B82F6.
Clue photos	In order uploaded by setter (1, 2, 3)	Thumbnail strip below main image. Border colour #8B5CF6.
Text clues	In order selected/written by setter (1, 2, 3)	Pill card below images. Background #F59E0B11, amber text.

The guesser sees available reveal types grouped by category with counts (e.g. “Sub-portion (2 available)”). They choose the category; the next item in that category is revealed. They can mix categories freely.

Important: The initial image (first sub-portion or full photo) is always free and not counted as a reveal. The reveal count only includes additional pieces of information beyond the initial view.